

I 304

Fall 2025



TEXAS
The University of Texas at Austin

Course Title

PROGRAMMING FOR INFORMATICS

Course Information

Instructional Mode: Face-to-face

Meeting Times: Tue/Thu 05:00 PM - 06:30 PM

Meeting Location: UTC 1.116

Unique Number: 29555

Instructor

Huimin Xu

Email: huiminxu@utexas.edu

Office Hours and Location

Office Location: UTC 1.116

Office Hours: Right after class Tue/Thu.

Teaching Assistants

There are no teaching assistants in this class.

Welcome Message

This class offers an introduction to computer programming for those **without any prior knowledge or experience** in computer programming. If you already have programming experience, please explore options below for waiving the requirement to take this class.

Requirement to complete I304

This class is required for the undergraduate B.S. Informatics degree (see concentrations) and as a prerequisite for various other informatics courses. Alternative ways to meet this requirement include:

Complete this class (see Past & Future Offerings of this Course)

- Complete a suitable substitute course
 - BME 303 Introduction to Computing
 - CS 303E Elements of Computers and Programming
 - CS 312 Introduction to Programming
 - J 326C Introduction to Coding for Journalists
 - LIN 313 Language and Computers
 - LA 319 Computer Programming: Liberal Arts
 - MIS/BAX 304: Introduction to Problem Solving and Programming
 - SDS 322 Introduction to Scientific Programming
- Transfer credit, e.g., from AP Test
 - <https://utdirect.utexas.edu/ctl/cbe/petition/index.WBX>
- Test out of the requirement
 - Exam in Computer Science 303E
 - Exam in Computer Science 312

Catalog Description

Examine the fundamentals of computer programming and practice developing and documenting code. Restricted to informatics majors and students pursuing the informatics minor.

Overview of the Class

All instructions, assignments, readings, rubrics and essential information will be on the Canvas website at utexas.instructure.com. Check Canvas regularly. **Changes** to the schedule may be made at my discretion if circumstances require. I will announce any such changes in class and will also communicate them via a Canvas announcement. It is your responsibility to note these changes when announced, and I will do my best to ensure that you are notified of changes with as much advance notice as possible.

Weekly Class Topics and Activities

Week	Date	Class Topic	Assignments Due
1.	Aug 26/28	Syllabus Introduction to Python and Google Colab	
2.	Sep 2/4	Variables, expressions, and statements Conditional Execution, Functions, Loops and Iterations	
3.	Sep 9/11	Lists, Dictionaries, Tuples Dataframes	
4.	Sep 16/18	Question and hypothesis	
5.	Sep 23/25	Presentations with your own dataset	Sep 23
6.	Sep 30/Oct 2	Regression	
7.	Oct 7/9	Visualizaton	
8.	Oct 14/16	Machine Learning	
9.	Oct 21/23	Machine Learning	
10.	Oct 28/30	Group project discussion	Oct 30
11.	Nov 4/6	Working with ChatGPT	
12.	Nov 11/3	Working with ChatGPT	
13.	Nov 18/20	No class Nov 18 (Attending Conference) Q & A for final project	

Week	Date	Class Topic	Assignments Due
14.	Dec 2/4	Presentations - final project	Dec 2

Learning Outcomes

At the conclusion of this course, students will be able to:

- Have the skills required to solve problems by creating and modifying programs in Python.
- Have the knowledge of basic programming concepts (e.g. computational thinking, architecture, flowcharts, documentation, debugging, version control), their appropriate usage, and how and where to learn more
- Have the skills to communicate about programs and the programming process to help clarify their ideas and to communicate these ideas to others
- Have an attitude of confidence when reading, writing, or discussing computer code

Grading Policy

Grade breakdown for class

Grade	Cutoff
A	94%
A-	90%
B+	87%
B	84%
B-	80%
C+	77%
C	74%
C-	70%
D+	67%
D	64%
D-	60%
F	<60%

Overview of all Major Course Requirements and

Assignments

The following table represents how you will demonstrate your learning and how we will assess the degree to which you have done so.

Assessment Overview

Assignments	Points Possible	Percent of Total Grade
1. Individual presentation with one dataset	20	20
2. Group project	20	20
3. Individual presentation for final project	20	20
4. Individual report for final project	20	20
5. Attendance	20	20

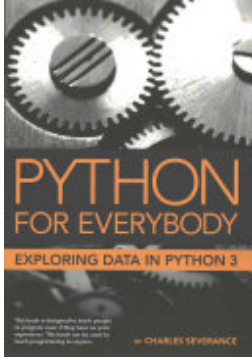
Attendance Policy

- No attendance will be taken the first week of class (since new people often enroll late)
- You may miss up to 4 classes with no grade penalty. No explanations are needed.
- Each unexcused, non-emergency absence after your allotted 4 absences will reduce the final grade
- Additional classes may be missed without grade penalty only for university-excused absences (e.g., religious holy days, jury duty, military service, officially documented illness or family loss ,etc.) with formal documentation provided by Student Outreach and Support(SOS).

Canvas Assignments

Due Date	Assignment Name	Points
	Aug 28	0

Required Course Materials



Python for Everybody
ISBN: 9781530051120
Authors: Charles R. Severance
Publication Date: 2016-04-09

Recommended Course Materials

Python Data Science Handbook

Read the book in its entirety online at <https://jakevdp.github.io/PythonDataScienceHandbook/>

Software

Software Python Development assignments will be completed using Google Colab, an approved browser-based development environment that allows students to write and run fully functional Python programs online. This eliminates the need to install Python on personal computers, ensuring all students use a consistent and suitable development platform.

Hardware

Students will need laptops able to run Firefox or other modern web browsers. iPads are NOT recommended for coding. If you must use an iPad, I highly recommend purchasing an external keyboard. Or, since iPad accessories are already pretty expensive, purchase a \$150 Chromebook. Programmers need keyboards. University-owned machines may be available for the times you will need to record your screen if your machine is incapable of doing so. Contact me early if you'd like help securing access to hardware you don't have or can't afford.

University Policies and Resources for Students Canvas Page

This Canvas [page](#) is a supplement to all UT syllabi and contains University policies and resources that you can refer to as you engage with and navigate your courses and the university.

Counseling and Mental Health Center (CMHC)

Students who are struggling for any reason and who believe that it might impact their performance in the course are urged to reach out to Bryce Moffett if they feel comfortable. This will allow her to provide any resources or accommodations that she can. If immediate mental health assistance is needed, call the Counseling and Mental Health Center (CMHC) at 512-471-3515 or you may also contact Bryce Moffett, LCSW (iSchool CARE counselor) at 512-232-4449. Bryce's office is located in FAC18S and she holds drop in Office Hours on Wednesday from 2-3pm. For urgent mental health concerns, please contact the CMHC 24/7 Crisis Line at 512-471-2255.

Important Safety Information

Carrying of Handguns on Campus

Students in this class should be aware of the following university policies related to Texas' Open Carry Law:

- Students in this class who hold a license to carry are asked to [review the university policy regarding campus carry](#).
- Individuals who hold a license to carry are eligible to carry a concealed handgun on campus, including in most outdoor areas, buildings and spaces that are accessible to the public, and in classrooms.
- It is the responsibility of concealed-carry license holders to carry their handguns on or about their person at all times while on campus. Open carry is NOT permitted, meaning that a license holder may not carry a partially or wholly visible handgun on campus premises or on any university driveway, street, sidewalk or walkway, parking lot, parking garage, or other parking area.

A Notice of Academic Accommodations from Disability and Access (D&A)

Accessible/Compliant Statement:

If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). You may refer to D&A's website for contact and more information: <http://community.utexas.edu/disability/>. If you are already registered with D&A, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations.

Accessible, Inclusive, and Compliant Statement:

The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). Please refer to D&A's website for contact and more information: <http://community.utexas.edu/disability/>. If you are already registered with D&A , please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.